Started on	Wednesday, 7 October 2026, 9:41 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 9:41 PM
Time taken	26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    // Phase 1: Find intersection point
    int slow = a[0];
    int fast = a[0];

    do {
        slow = a[slow];
        fast = a[a[fast]];
    } while (slow != fast);

    // Phase 2: Find duplicate number
    slow = a[0];

    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }

    printf("%d", slow);

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Friday, 9 October 2026, 3:03 PM
State	Finished
Completed on	Friday, 9 October 2026, 3:03 PM
Time taken	32 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, i, j;
    scanf("%d", &n);

    int arr[n];

    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    for (i = 0; i < n; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                printf("%d\n", arr[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Friday, 9 October 2026, 3:04 PM
State	Finished
Completed on	Friday, 9 October 2026, 3:05 PM
Time taken	26 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int N1, N2, i = 0, j = 0;

        scanf("%d", &N1);
        int a[N1];

        for (int k = 0; k < N1; k++) {
            scanf("%d", &a[k]);
        }

        scanf("%d", &N2);
        int b[N2];

        for (int k = 0; k < N2; k++) {
            scanf("%d", &b[k]);
```

```
        }

        int first = 1;

        while (i < N1 && j < N2) {
            if (a[i] == b[j]) {
                if (!first) {
                    printf(" ");
                }
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            } else if (a[i] < b[j]) {
                i++;
            } else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Friday, 9 October 2026, 3:06 PM
State	Finished
Completed on	Friday, 9 October 2026, 3:06 PM
Time taken	28 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array
2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int T;
    scanf("%d", &T);

    while (T--) {
        int n1, n2;
        scanf("%d", &n1);

        int a[n1];
        for (int i = 0; i < n1; i++) {
            scanf("%d", &a[i]);
        }

        scanf("%d", &n2);

        int b[n2];
        for (int i = 0; i < n2; i++) {
            scanf("%d", &b[i]);
        }
```

```
        int i = 0, j = 0, first = 1;

        while (i < n1 && j < n2) {
            if (a[i] == b[j]) {
                if (!first) {
                    printf(" ");
                }
                printf("%d", a[i]);
                first = 0;
                i++;
                j++;
            } else if (a[i] < b[j]) {
                i++;
            } else {
                j++;
            }
        }

        printf("\n");
    }

    return 0;
}
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Friday, 9 October 2026, 3:08 PM
State	Finished
Completed on	Friday, 9 October 2026, 3:09 PM
Time taken	47 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n) {
        if (i != j && a[j] - a[i] == k) {
            printf("1\n");
            return 0;
        } else if (a[j] - a[i] < k) {
            j++;
        } else {
            i++;
        }

        if (i == j) {
            j++;
        }
    }

    printf("0\n");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Friday, 9 October 2026, 3:11 PM
State	Finished
Completed on	Friday, 9 October 2026, 3:11 PM
Time taken	27 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int main() {
    int n, k;
    scanf("%d", &n);

    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    scanf("%d", &k);

    int i = 0, j = 1;

    while (i < n && j < n) {
        int diff = a[j] - a[i];

        if (i != j && diff == k) {
            printf("1\n");
            return 0;
        }

        if (diff < k) {
            j++;
        } else {
            i++;
        }

        if (i == j) {
            j++;
        }
    }

    printf("0\n");

    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Started on	Wednesday, 7 October 2026, 10:49 PM
State	Finished
Completed on	Wednesday, 7 October 2026, 10:50 PM
Time taken	55 secs
Marks	2.00/2.00
Grade	**20.00** out of 20.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Write a C program to find the Nth term in the fibonacci series.

For example:

Input	Result
0	0
1	1
4	3

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n, a = 0, b = 1, c, i;

    scanf("%d", &n);

    if (n == 0) {
        printf("0");
    } else if (n == 1) {
        printf("1");
    } else {
        for (i = 2; i <= n; i++) {
            c = a + b;
            a = b;
            b = c;
        }
        printf("%d", b);
    }

    return 0;
}
```

	Input	Expected	Got	
✔	0	0	0	✔
✔	1	1	1	✔
✔	4	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Question 2 | Correct Mark 1.00 out of 1.00

Given two numbers, write a C program to swap the given numbers.

For example:

Input	Result
10 20	20 10

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int a, b, temp;

    scanf("%d %d", &a, &b);

    temp = a;
    a = b;
    b = temp;

    printf("%d %d", a, b);

    return 0;
}
```

	Input	Expected	Got	
✔	10 20	20 10	20 10	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.